

A bilinear Laguerre endpoint test

Research note; independently reviewed

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This note isolates the algebraic improvement from the preliminary $\Omega(n^{5/4})$ per-die bound to $\Omega(n^{4/3})$. The complete comparison-tensor proof is in `../../power_per_die_lower_bound.tex`.

Lemma 1 (Bilinear extension). *Suppose two linear functionals F, G on polynomials satisfy*

$$|F(Q^2) - G(Q^2)| \leq C\|c\|_1^2\delta \quad \text{whenever } Q(z) = \sum_{j=0}^d c_j L_j(\theta z).$$

If Q_1, Q_2 have coefficient sums B_1, B_2 , respectively, then

$$|F(Q_1 Q_2) - G(Q_1 Q_2)| \leq 2CB_1 B_2 \delta.$$

Proof. The case $B_1 B_2 = 0$ is immediate. Otherwise set $a = \sqrt{B_2/B_1}$ and use

$$Q_1 Q_2 = \frac{1}{4}[(aQ_1 + a^{-1}Q_2)^2 - (aQ_1 - a^{-1}Q_2)^2].$$

Each square has coefficient sum at most $2\sqrt{B_1 B_2}$. □

Lemma 2 (The tapered endpoint test has small bilinear norm). *Let*

$$T_d(y) = \sum_{j=0}^{d-1} (1 - j/d) L_j(y), \quad R_d(z) = \left(\frac{2}{\theta d} - z \right) T_d(\theta z)^2.$$

Then $R_d = Q_d S_d$, where $Q_d = T_d(\theta z)$ has coefficient sum $(d+1)/2$, and S_d has coefficient sum at most $2/\theta$.

Proof. The Laguerre three-term recurrence gives the exact identity

$$yT_d(y) = \frac{1}{d} \sum_{j=0}^{d-1} L_j(y) - L_d(y).$$

Hence $S_d = (2/(\theta d) - z)T_d(\theta z)$ has coefficients

$$[L_j(\theta z)]S_d = \begin{cases} (1 - 2j/d)/(\theta d), & 0 \leq j < d, \\ 1/\theta, & j = d. \end{cases}$$

Their absolute sum is at most $2/\theta$. □

The scalar endpoint comparison in the complete proof gives $\delta = d^2/m + e^{-cm}$ uniformly for $d \leq m^{1/3}$: its summation requires only $d/\sqrt{m} \rightarrow 0$, so the earlier provisional range $d \leq m^{1/4}$ was unnecessary. The two lemmas therefore give

$$|\mathbb{E}_\mu R_d - \beta_\ell(R_d)| \leq C(d^3/m + de^{-cm}).$$

The positive scalar expectation is bounded below by an absolute constant. Choose $d = \lfloor \kappa m^{1/3} \rfloor$ with a sufficiently small fixed $\kappa > 0$. The endpoint-forcing error is then small, while the normalized Christoffel-square error is $O(d^2/m) = O(1/d)$. The remaining endpoint atom argument is unchanged, and gives $w_K, w_1 = O(m^{-4/3})$, hence every equiprobable die has $\Omega(n^{4/3})$ faces.